CSE 242 Assignment 1

Wednesday, October 7, 2026 9:14 PM

Q1. $A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$

$$AB = \begin{bmatrix} 3+0 & 6-4 \\ 2+0 & 4-1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

$$BA = \begin{bmatrix} 3+4 & 4+2 \\ 0-2 & 0-1 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ -2 & -1 \end{bmatrix}$$

$$(A+B)^T = \begin{bmatrix} 3+1 & 4+2 \\ 2+0 & 1-1 \end{bmatrix}^T = \begin{bmatrix} 4 & 6 \\ 2 & 0 \end{bmatrix}^T = \begin{bmatrix} 4 & 2 \\ 6 & 0 \end{bmatrix}$$

$$AB^T = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}^T = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

$\text{trace}(A) = 3+1 = 4$

$\text{trace}(AB) = 3+3 = 6$

Q2. $M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

$\det(\lambda I - M)$

$$\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} \lambda-2 & -1 \\ -1 & \lambda-2 \end{bmatrix}$$

$$\det \begin{bmatrix} \lambda-2 & -1 \\ -1 & \lambda-2 \end{bmatrix} = (\lambda-2)^2 - (1) = 0$$

$(\lambda-2)^2 = 1$

$\lambda - 2 = \pm 1$

$\lambda = 3 \quad \lambda = 1$

$$\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 3-2 & 0+1 \\ 0+1 & 3-2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x+y \\ x+y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$x = -y$

$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1-2 & 0+1 \\ 0+1 & 1-2 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -x+y \\ x-y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-x+y = 0 \quad x-y = 0$

$x = y$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Q3. Prove trace(AB) = trace(BA)

let A be a matrix of $i \times j$ dimension and B be a matrix of $j \times i$ dimension, without loss of generality.

A can be represented as: $\begin{bmatrix} x_{11} & \cdots & x_{i1} \\ \vdots & & \vdots \\ x_{1j} & \cdots & x_{ij} \end{bmatrix}$

and

B as: $\begin{bmatrix} y_{11} & \cdots & y_{j1} \\ \vdots & & \vdots \\ y_{1i} & \cdots & y_{ji} \end{bmatrix}$ $\quad x_{11}y_{11} + \ldots + x_{i1}y_{1i}$

$\quad y_{11}x_{11} + \ldots + y_{j1}x_{ij}$

By matrix multiplication, $AB_{ij} = \sum_{k=1}^{n} x_{ik}y_{ij}$ and $BA_{ji} = \sum_{k=0}^{n} y_{jk}x_{ki}$

$$\text{trace}(AB) = \sum_{i=1}^{n} (AB)_{ii} = \sum_{i=1}^{n} \sum_{j=1}^{m} x_{ik}y_{ki}$$

$$\text{trace}(BA) = \sum_{k=1}^{n} (BA)_{kk} = \sum_{k=1}^{m} \sum_{i=1}^{n} x_{ki}y_{ik} = \sum_{i=1}^{n} \sum_{k=1}^{m} x_{ik}y_{ki}$$

thus trace(AB) = trace(BA)

Q4. Find the inverse of the matrix $C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$

$$\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\frac{1}{\det(C)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{24-14} \begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 3/5 & -7/10 \\ -1/5 & 2/5 \end{pmatrix} = \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

Q1. $P(X=k) = c \cdot \frac{3^k}{k!} \quad k = 0, 1, 2 \ldots$

a. Find the normalization constant c

$P(X=k) = c \cdot \frac{3^k}{k!}$

Following the Poisson probability mass function, $P(X=k) = \frac{\lambda^k}{k!} \cdot e^{-\lambda}$, $c = e^{-k}$ and $\lambda = 3$ in this case, so $c = e^{-3}$.

b. Following the Poisson PMF, $\mathbb{E}[X] = \lambda = 3$ and $Var(X) = \lambda = 3$

Q2. Given the two events A and B with $P(A) = 0.5$, $P(B) = 0.3$, $P(A|B) = 0.6$

a. $P(A \cap B) = P(A|B) \cdot P(B) = 0.6 \cdot 0.3 = 0.18$

b. $P(B|A) = P(B \cap A) / P(A) = 0.18 / 0.5 = 0.36$

c. $P(A \cup B) = P(A) + P(B) - P(B \cap A) = 0.5 + 0.3 - 0.36 = 0.8 - 0.36 = 0.44$

Q3. Suppose you have a biased coin with unknown probability of heads $\theta = P(H)$ you flip this coin 5 times and observe the sequence: H, H, T, H, T

a. Derive the MLE for θ

$L(\theta) = P(H)P(H)P(T)P(H)P(T)$

$P(H) = \theta \quad P(T) = 1-\theta$

$L(\theta) = (\theta)^3(1-\theta)^2$

$\ell(\theta) = \ln(\theta^3(1-\theta)^2) = 3\ln\theta + 2\ln(1-\theta)$

$\frac{\partial \ell}{\partial \theta} = \frac{3}{\theta} - \frac{2}{1-\theta} = 0$

$3(1-\theta) - 2(\theta) = 0$

$3 - 3\theta - 2\theta = 0$

$3 = 5\theta$

$\theta = \frac{3}{5} = 0.6$

b. compute the MAP estimate for θ, assuming a prior distribution

$P(\theta) = \{2\theta,\ 0 \le \theta \le 1,\ 0$ otherwise

$P(D|\theta) = \theta^3(1-\theta)^2$

$P(\theta|D) = P(D|\theta)P(\theta)$

$P(\theta|D) = (\theta^3(1-\theta)^2)(2\theta) = (2\theta^4(1-\theta)^2)$

$\ell(\theta) = \ln(2\theta^4(1-\theta)^2) = \ln 2 + 4\ln\theta + 2\ln(1-\theta)$

$\frac{\partial \ell}{\partial \theta} = \frac{4}{\theta} - \frac{2}{1-\theta}$

$4(1-\theta) - 2\theta = 0$

$4 - 4\theta - 2\theta = 0$

$4 - 6\theta = 0 \rightarrow \theta = \frac{4}{6} = \frac{2}{3}$

Q4. consider a Gaussian distribution with parameters μ (mean) and σ^2 (variance). Given the dataset $X = 1, 3, 5, 7$, derive the MLE for μ and σ^2.

$$f(x|\mu,\sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$L(\mu,\sigma^2) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

$$= \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n e^{\sum_{i=1}^{n} \frac{-(x_i-\mu)^2}{2\sigma^2}} = \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2}$$

$$= \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2}$$

$$= (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2}$$

$$\ell(\mu,\sigma^2) = \ln\left((2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2}\right)$$

$$= -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\ \cancel{\ln(e)}\ 1$$

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu)$$

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = 0$$

$$\sum_{i=1}^{n} x_i - n\mu = 0$$

$$\sum_{i=1}^{n} x_i = n\mu$$

$$(1+3+5+7) = 4\mu$$

$$\underbrace{\mu = \frac{1+3+5+7}{4} = \frac{16}{4} = 4}_{\text{MLE } \mu}$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i-\mu)^2$$

$$\underbrace{\begin{aligned} -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i-\mu)^2 &= 0 \\ \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i-\mu)^2 &= \frac{n}{2\sigma^2} \\ \sum_{i=1}^{n}(x_i-\mu)^2 &= n\sigma^2 \\ \sigma^2 &= \sum_{i=1}^{n}(x_i-\mu)^2 \cdot \frac{1}{n} \\ \sigma^2 &= \frac{(1-4)^2+(3-4)^2+(5-4)^2+(7-4)^2}{4} = \frac{(-3)^2+(-1)^2+(1)^2+(3)^2}{4} \\ \sigma^2 &= \frac{9+1+1+9}{4} = \frac{20}{4} = 5 \end{aligned}}_{\text{MLE } \sigma^2}$$

Q5. You observe examples drawn from a Gaussian distribution with unknown mean μ and known variance $\sigma^2 = 4$. Given a Gaussian prior for the mean μ:

$\mu \sim N(0,1)$

derive the MAP estimate of μ after observing a single data point $x=5$

$$f(x|\mu,\sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$P(x=5|\mu) = \frac{1}{\sqrt{8\pi}} e^{-\frac{(5-\mu)^2}{8}}$$

$$P(\mu) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(\mu-0)^2}{2}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{\mu^2}{2}}$$

$$P(\mu|x=5) = P(x=5|\mu)P(\mu) = \left(\frac{1}{\sqrt{8\pi}} e^{-\frac{(5-\mu)^2}{8}}\right)\left(\frac{1}{\sqrt{2\pi}} e^{-\frac{\mu^2}{2}}\right) = \frac{1}{4\pi} e^{-\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2}}$$

$$\ell(\mu) = \ln\left(\frac{1}{4\pi}\right) - \frac{(5-\mu)^2}{8} - \frac{\mu^2}{2}\ \cancel{\ln(e)}\ 1$$

$$\frac{\partial \ell}{\partial \mu} = \frac{5-\mu}{4} - \mu$$

$$\frac{5-\mu}{4} - \mu = 0 \rightarrow 5 - \mu = 4\mu \rightarrow 5 = 5\mu$$

$\mu = 1$

Q6. Prove $Var(X) = E(X^2) - [E(X)]^2$

$$Var(X) = E[(X - E[X])^2] = E[X^2 - 2XE[X] + (E[X])^2]$$

$$= E[X^2] - 2E[X]E[X] + (E[X])^2$$

$$= E[X^2] - (E[X])^2 - \cancel{(E[X])^2} + \cancel{(E[X])^2}$$

$$= E[X^2] - (E[X])^2$$

Q1. derive $Var(u^T x) = u^T \Sigma u$

$$Var(u^T x) = E[(u^T x - E[u^T x])^2]$$

because u is a unit vector, it can be factored outside of the expectation so

$E[u^T x] = u^T E[X]$

so $Var(u^T x) = E[(u^T x - u^T E[X])^2] = E[(u^T(x - E[X]))^2]$

$= E[(u^T(x - E[X]))(u^T(x - E[X]))]$

$u^T(x - E[X])$ is a scalar because it take the dot product between two vectors $\rightarrow$ therefore

$u^T(x - E[X]) = (u^T(x - E[X]))^T$

$E[(u^T(x - E[X]))((u^T(x - E[X]))^T)] = E[(u^T(x - E[X]))((x - E[X])^T u)]$ by $(AB)^T = B^T A^T$

$= u^T E[(x - E[X])(x - E[X]^T)] u = u^T E[(x - E[X])(x - E[X]^T)] u$

by defining $\Sigma = E[(x - E[X])(x - E[X]^T)]$, we get $Var(u^T x) = u^T \Sigma u$